Bryan Zhao

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EDUCATION

University of Southern California **Expected Jun. 2030**
B.S. Computer Science + Physics
- Relevant Coursework: PHYS 161, MATH 226, EE155
- Activities and Societies: Sigma Eta Pi (SEP), TroyLabs, Liquid Propulsion Lab (LPL), Chinese-American Student Association (CASA)

Saratoga High School **Jun. 2026**
High School Diploma *Saratoga, CA*
- Academic GPA: 3.9 UW/4.5 W; SAT: 1550, 6x Scholar of Distinction, Very High Honors
- Relevant Coursework: AP Physics C: E&M+Mech, AP Calculus BC, AP Lang, AP Chemistry, AP US History

Dual Enrollment
- Relevant Coursework: Elementary Statistics, Intermediate Calculus, Linear Algebra

EXPERIENCE

Saratoga High School VEX Robotics Program **Aug. 2022 – Jun. 2026**
Co-Founder, Co-President & Team Captain *Saratoga, CA*
- Scaled organization to 50+ engineers with 100+ annual applicants; secured $225,000+ via sponsorships and tournament operations
- Architected autonomous navigation and control systems across 12+ robots, implementing particle-filter localization, sensor alignment, and custom odometry-based path tracking algorithms
- Designed power-takeoff transmissions and various mechanisms in CAD; performed stress simulations and operated CNC/lathe machinery to machine high-load components.
- Spearheaded international advocacy campaign for equity, authoring an open letter (60,000+ impressions, 4,000+ verified signatures)

University of California, Santa Cruz **May 2024 – Feb. 2026**
Research Intern *Santa Cruz, CA*
- Engineered edge-to-cloud multimodal pipeline integrating Raspberry Pi, ambient sensors, and Android agents; captured synchronized video, audio, and sensor telemetry to detect senior falls and generate adaptive care plans.
- Optimized video-language models (VLMs) for edge deployment, improving computational and energy efficiency by 40% while increasing action classification accuracy by 4.6%.
- Designed evaluation benchmarks and data preprocessing pipelines across multimodal datasets; co-authored research paper accepted at NeurIPS 2024 Workshop on Video-Language Models.
- Selected for competitive engineering cohort (9% acceptance); conducted sentiment analysis and prototyped digital journaling tools to assess adolescent well-being.

Stanford Institute for Human-centered Artificial Intelligence (HAI) **Jun. 2023 – Jul. 2023**
Program Intern @ AI4ALL *Remote*
- Collaborated with Stanford PhD candidates to model and optimize aircraft taxiing systems using state estimation for PID controllers.
- Lectures from Prof. Fei-Fei Li and leading AI experts (4–6% worldwide acceptance rate)

LEADERSHIP

Saratoga Associated Student Body (ASB) **Jun. 2023 – Jun. 2026**
Associated Student Body Officer *Saratoga, CA*
- 1 of 2 ASB Club Commissioners and 1 of 6 ASB Officers, managing 120+ student organizations and 1000+ students.
- Led 2026 Prom planning ($80,000+ budget) and coordinated educational, sporting, and cultural events throughout the year

City Of Saratoga **Jun. 2022 – Jun. 2024**
Youth Commissioner *Saratoga, CA*
- Appointed by City Council as 1 of 11 students citywide; led design for a $150,000 eco-park for environmental awareness

AWARDS/HONORS

- 2022 + 2025 VEX Robotics World Semifinalist Alliance Captain - 4/20,000+ Teams (Top 0.02% in the World)
- Paper accepted in NeurIPS 2024 Workshop on Video-Language Models
- 2024 VEX Robotics U.S. Open Excellence Award (Highest Team Honor) - 1/2000+ Teams, Top 0.05% in the U.S.